Proof. If (P, f) were polyhedrally sofic, then its dual rooted ADT would be a Benjamini–Schramm limit of the rooted marked dual triangulations of uniformly face-rooted finite trivalent hyperbolic polyhedra. Theorem 4.3 would then imply that P is parabolic almost surely, contradicting

$$\mathbb{P}(P \text{ is hyperbolic}) > 0.$$

□

It is natural to ask whether the converse also holds.

Conjecture 4.5. *Let (P, f) be an ergodic, tame, unimodular random rooted infinite trivalent hyperbolic polyhedron whose dual ADT is almost surely infinite, locally finite, simple, one-ended, and satisfies* (Z1)–(Z4)*. Then*

$$P \text{ is almost surely parabolic} \quad \Longrightarrow \quad (P, f) \text{ is polyhedrally sofic.}$$

This does not contradict the Aldous–Lyons conjecture, since polyhedral soficity is substantially stronger than ordinary soficity: the finite approximations must arise from trivalent hyperbolic polyhedra and must preserve the dihedral angles. Thus

$$\text{polyhedrally sofic} \quad \Longrightarrow \quad \text{sofic},$$

but not conversely. Indeed, the underlying graph of a hyperbolic unimodular THP may still be approximable by triangulations of higher-genus surfaces. For a triangulation with N vertices on a closed orientable surface of genus g,

$$\frac{1}{N}\sum_{v} \deg(v) = 6 + \frac{12(g-1)}{N},$$

so such approximations are not subject to the spherical Gauss-Bonnet obstruction.

5. Refined ring lemma for RCP

Ring lemmas are a fundamental tool in circle packing theory. They give quantitative control on the ratios of neighboring radii and are crucial for convergence and random-walk estimates. In the classical circle packing setting, such estimates go back to the work of Rodin-Sullivan and have been used systematically in the study of infinite packings and discrete conformal geometry [35, 46, 52].

In this section we will prove the refined ring lemma for RCPs. Compared with the local ring lemma, we quantify the radius ratio by degrees: for a vertex v with neighbors v_i and $d_i = \deg(v_i)$, the ratio is controlled by an exponential factor in $\sum_i d_i$. Compared with the uniform ring lemma, we weaken the conditions by removing bounded degree assumption.

Lemma 5.1 (Refined ring lemma for RCP)**.** *Let (G, Θ) be a tame ADT satisfying (Z1)–(Z4), and assume that (Z2) is strengthened to (Z2′). Let $r : V \to (0, \infty)$ denote the radius function of an embedding in the plane. Then there exists a constant $C = C(\varepsilon_0, \varepsilon_1)$, where ε_0 is the constant in (Z2′) and ε_1 is the tameness constant in the angle upper bound $\Theta(e) \le \pi - \varepsilon_1$, such that for every edge $u \sim v$,*

$$\frac{r(v)}{r(u)} > e^{-C \cdot S(u)},$$

where $S(u) = \sum_{v' \sim u} \deg(v')$ is the flower degree *of u.*